Tianxiao He

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EDUCATION

Columbia University — New York, NY
B.S., Computer Science — May 2023
Research Advisor: Liam Paninski
GPA: 3.9/4.0 (*cum laude*)

New York University — New York, NY
Ph.D., Computer Science — Sep 2023 - Present
Research Advisor: Erdem Varol

PUBLICATIONS

T. He, S. Harvey, A. Williams. "How Data Augmentation Shapes Neural Representations". **NeurIPS 2026;** ICLR 2026 ReAlign Workshop (**Oral**)

T. He*, M. Patel*, C. Li, A. Maslarova, M. Vöröslakos, N. Ramanathan, W. Hung, G. Buzsáki, and E. Varol. "Self supervised learning for in vivo localization of microelectrode arrays using raw local field potential. " **NeurIPS 2025**

Y. Zhang*, **T. He***, J. Boussard, C. Windolf, O. Winter, E. M. Trautmann, N. Roth, H. Barrel, M. M. Churchland, N. Steinmetz, E. Varol, C. L. Hurwitz, and L. Paninski. "Bypassing spike sorting: Density-based decoding using spike localization from dense multielectrode probes." **NeurIPS 2023 (spotlight)**

J. Adhinarta, J. Dong, **T. He**, J. Huang, D. Sprague, J. Wan, H. J. Lee, Z. Yu, H. Lu, E. Yemini, S. Kato, E. Varol, and D. Wei. "WormID-Benchmark: Extracting Whole-Brain Neural Dynamics of C. elegans at the Neuron Resolution". bioRxiv 2025

INTERNSHIP

Research Scientist Intern at Center for Computational Neuroscience
Flatiron Institute — May 2025 – August 2025

- Developed a geometric framework to quantify how data augmentation interacts and reshapes neural representations
- Embedded hidden representations from CNNs and ViTs into a smooth manifold using shape analysis invariant to scale, rotation and reflection
- Demonstrated consistent and stereotyped trajectories across augmentation strength and type, enabling more effective model ensembling (accepted to NeurIPS 2026)

Research Trainee in Computational & Cognitive Neuroscience Program
Cold Spring Harbor Asia — June 2023 – July 2023

- Developed a brain-inspired model of dorsal and ventral pathways in visual processing
- Implemented a recurrent neural network with continual learning across sequential tasks
- Demonstrated compositional and flexible task representations within the model, enabling learning without catastrophic forgetting

TECHNICAL SKILLS

- **Programming**: Python, Java, Git, SQL, PyTorch, TensorFlow, Hugging Face
- **Machine Learning**: Self supervised learning, Time Series Foundation models
- **Other skills**: Signal Processing, Neural Signal Decoding, Vocalization Segmentation

CONFERENCE PRESENTATION & WORKSHOP

Decoding Anatomical Structure from Neural Population Activity using Self-supervised Learning (**COSYNE 2026**)

Self supervised learning for in vivo localization of microelectrode arrays (**SAND 2025**)

Blind in vivo localization of microelectrode arrays via functional correlation patterns in the mouse hippocampus (**SFN 2024)**

Density-based Neural Decoding using Spike Localization for Neuropixels Recordings (**COSYNE 2023)**

PROFESSIONAL SERVICE

Conference Reviewer
NeurIPS 2023, NeurIPS 2025, MICCAI 2025, NeurIPS 2026 (Top Reviewer) & NeurIPS 2026 GenAI4Health workshop

Program Committee
NeurIPS 2025, 2026 Brain & Body Foundation Model Workshop

Journal Reviewer
PLOS Computational Biology

Teaching Experience
Guest Instructor, Neuroinformatics Class, New York University

AWARDS & FELLOWSHIPS

NYU School of Engineering PhD Fellowship	2023-2024
Columbia University Dean's List	2021-2022
Simon’s Rock Dean's List	2019-2021
Simon’s Rock Merit Scholarship	2018-2021